

SONAL BHATIA

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Education

Princeton University

Sep 2022 - Jan 2027

B.S.E. Mechanical & Aerospace Engineering, Minors in Computer Science and Robotics

Princeton, NJ

- **Relevant Coursework:** Computer Vision, Robotics, Algorithms & Data Structures, Computer Architecture, Microprocessors, Operating Systems, Distributed Systems, Automatic Control Systems, Optimal Control, Space System Design, Aircraft Flight Dynamics, Ethics of Computing, High-Tech Entrepreneurship.
- **Senior Thesis:** *Multi-Particle Control for Nonlinear Data Assimilation in Chaotic Systems*, an imitation-learning framework for amortizing optimal particle-filter nudging in chaotic dynamical systems, achieving a **77% RMSE reduction** on Lorenz-63 and scaling to Lorenz-96; advised by Prof. Ryne Beeson (MAE 497, April 2026).
- **Extracurricular:** Rocketry Club (Co-President; managed 7 project teams across 95+ members), Autonomous Vehicles Engineering, Society of Women Engineers, Rewriting the Code Fellow, Reboot Representation Scholar, Students for the Exploration and Development of Space, American Institute of Aeronautics and Astronautics.

Cornell University

May 2024 - Apr 2025

Certificate in Machine Learning Foundations, Break Through Tech AI

Grade: 94%

- Completed a foundational ML curriculum covering supervised and unsupervised learning, neural network architectures, model evaluation, and deployment pipelines; culminated in a real-world industry challenge project.

Skills

Languages: Python, Java, C++, Go, Swift, Objective-C, Assembly, HTML, CSS, JavaScript, TypeScript.

GPU/ML: PyTorch, TensorFlow, MLX, Core ML, Metal Performance Shaders, CUDA, quantization (INT8, INT4, mixed-precision), LLM tooling, NumPyro, OpenCV, scikit-learn, Pandas, NumPy.

Quantitative: Linear Algebra, Bayesian Regression, Monte Carlo/MCMC, Kalman Filtering, Regression Discontinuity Design.

Systems & APIs: REST APIs, Node.js, React, SQL/NoSQL databases, Docker, Git, CI/CD, AWS/GCP.

Robotics & Systems: Robot Operating System (ROS), FastMCP, Vulkan, OpenGL, OpenVDB.

Tools & Hardware: Bootstrap, MATLAB, CAD (SolidWorks, Fusion 360, CREO), Raspberry Pi, Linux.

Experience

Apple Inc.

June – August 2026

Artificial Intelligence Performance Engineer Intern

Cupertino, CA

- Profiled and resolved **performance bottlenecks** across the stack via Thermal, Acoustics, Power, and Performance (TAPP) analysis, shipping fixes for 10 issues across graphics and ML **inference pipelines** on next-generation MacBook Pro and Mac Studio **system and SoC architectures**.
- Diagnosed a compute-bound bottleneck in a production **LLM MoE inference** pipeline; **built internal profiling tooling** that surfaced 3 additional issues, boosting performance by **30%**, and presented findings to stakeholders to drive engineering decisions.
- Profiled end-to-end model execution graphs to identify compute-bound and memory-bound stages, informing hardware co-design decisions for upcoming Apple Silicon releases.

Princeton University, Keller Center for Engineering Education

Feb – May 2026

Undergraduate Course Assistant – Linear Systems (EGR 154)

Princeton, NJ

- Led weekly office hours and problem-solving sessions for **100+ students**, reinforcing **linear algebra** and systems theory concepts including eigendecomposition, SVD, state-space representations, and frequency-domain analysis.
- Graded problem sets and provided detailed written feedback; collaborated with graduate TAs to address conceptual gaps identified across student submissions.

Apple Inc.

January – August 2025

Artificial Intelligence Engineer Intern

Beaverton, OR

- Engineered a 3D-GenAI post-processing pipeline from scratch, developing **parallel algorithms** that implement the **Dual Contouring**, **Catmull-Clark**, and **OpenVDB** papers as production modules, enabling on-device 3D quad-mesh generation up to **1024³** resolution.
- Built and deployed a model **quantization** toolkit (INT8, INT4, mixed-precision) across 8 GenAI models, reducing memory footprint by **44%** and unlocking on-device deployment; profiled and optimized **inference** across GPU architectures, increasing GPU utilization by **15%**.
- Presented quantization findings and profiling methodology to the broader systems team, directly informing the on-device inference strategy for the following release cycle.

The Estée Lauder Companies Inc.

August – December 2024

Machine Learning Engineer Intern

New York City, NY

- Designed and shipped a **Bayesian Regression** model trained on 50,000+ transactions into a targeting pipeline, improving targeting accuracy by **40%**; built a **Monte Carlo/MCMC** inference pipeline for a Regression Discontinuity Design, reducing credible interval widths by **15%**.
- Delivered credible interval estimates for gift-driven revenue lift to the analytics and marketing teams, directly informing budget allocation for the following promotional cycle.

Johns Hopkins University Applied Physics Laboratory

June – October 2024

Software Development Engineer Intern

Laurel, MD

- Built classified computer graphics software in **C++** for threat signature modeling; authored 75+ automated test cases, resolving cross-API rendering discrepancies between **OpenGL** and **Vulkan** backends by **80%**.
- Designed the software architecture for an upcoming rendering feature, producing a technical specification reviewed and approved by senior engineers.

Space Technology & Aeronautical Rocketry (STAR)

December 2023 – January 2024

Astrodynamics Intern & Team Lead

Remote

- Led a team of 5 interns** to establish viable lunar trajectory parameters via a trans-lunar injection maneuver, designing and validating the flight path using the General Mission Analysis Tool (GMAT) and custom C++ scripts.
- Optimized trajectory design by iterating over launch windows, Δv budgets, and insertion burn parameters, delivering a feasibility report with recommended mission profiles.

Princeton University, Department of Computer Science

Feb – Dec 2024

Undergraduate Course Assistant – Linear Systems & Intro to Computer Science

Princeton, NJ

- Led weekly office hours for **150+ students**, teaching applications of **linear algebra** (eigendecomposition, SVD, PCA) to CS problems including image compression and dimensionality reduction.
- Collaborated with faculty to design problem sets and exam questions, contributing to a measurable 5% improvement in the Linear Systems final exam average.

EMTech Space S.A.

June – August 2023

Space Software Engineer Intern

Athens, Greece

- Designed and implemented a novel **genetic algorithm**-based orbit design tool, cutting computation time by **30%** across 1,000+ scenarios; authored 3 technical reports approved by the European Space Agency for the Hellenic Space Dawn CubeSat mission.
- Conducted a mission analysis covering orbital mechanics, launch windows, ground station coverage, and link budgets.

Research

Self-Organizing Swarms & Robotics (SSR) Lab, Princeton University

Aug – Dec 2026

Undergraduate Researcher, PI: Prof. Radhika Nagpal

Princeton, NJ

- Building a fleet of 6-10 Tumblenauts microgravity inspection robots alongside a PhD student, developing motion planning and ML-based autonomy software ahead of a ZeroG parabolic flight test in November 2026.
- Designed and implemented tumbling-target motion planning for the Tumblenauts robots, generating stable approach and grasp trajectories under free-floating, unactuated tumble dynamics with no reliable gravity reference frame.

Beeson Group, Princeton University

Aug – Dec 2026

Undergraduate Researcher, Advisor: Prof. Ryne Beeson

Princeton, NJ

- Researching physics-informed neural network (PINN) methods for tropical cyclone vortex initialization, extending an existing forecasting framework.
- Training the PINN to jointly satisfy observational fit and the governing vortex dynamics equations, embedding physical constraints directly into the loss function to improve initialization accuracy over purely data-driven baselines.

Senior Thesis, Department of Mechanical & Aerospace Engineering, Robotics

Aug 2025 – April 2026

Advisor: Prof. Ryne Beeson; Reader: Prof. Christine Allen-Blanchette

Princeton, NJ

- Thesis: *Multi-Particle Control for Nonlinear Data Assimilation in Chaotic Systems* (189 pages, April 2026).
- Developed an imitation-learning framework that amortizes the cost of optimal particle-filter nudging offline: a Feynman-Kac Monte Carlo oracle generates expert controls across a low-discrepancy sample of the chaotic attractor, with Girsanov-Radon-Nikodym corrections applied along each trajectory to preserve importance-sampling validity.
- Trained a neural network via behavioral cloning to imitate the oracle's per-particle control policy, replacing expensive online Monte Carlo rollouts with a single forward pass at deployment; evaluated four model variants: a per-particle regression baseline, two DeepSets ensemble-aware controllers with weight-aware pooling, and a residual architecture initialized from the per-particle checkpoint.

- On Lorenz-63, the per-particle controller achieved a **77% reduction** in normalized RMSE relative to the uncontrolled particle filter and a 4× improvement over the oracle-nudged particle filter, at 70% lower computational cost, explained by an implicit Girsanov regularization effect from Huber regression on the heavy-tailed oracle distribution.
- Demonstrated scaling to Lorenz-96 ($K = 8$ state variables) with no architectural changes, matching oracle accuracy while maintaining higher normalized effective sample size at 27% lower cost; argued that importance-sampling correctness, not model complexity, is the primary constraint on learned particle filter controllers.

Junior Independent Work

May 2024 – Jan 2026

Advisor: Prof. Ryne Beeson; Readers: Profs. Kolemen & Rowley

Princeton, NJ

- IW F25: *Learning Particle-Wise Control for Nonlinear Data Assimilation: From Diffusion Models to Supervised Regression* (86 pages, January 2026).
- Conducted a systematic comparison of diffusion-based and regression-based learned control for particle filtering, identifying a structural mismatch between generative models and importance sampling: diffusion models violate particle independence, lack importance-weight awareness, and are locally inconsistent on the chaotic attractor.
- Built a supervised regression framework that learns a deterministic mapping from particle state, observation, and time-to-observation to HJB-inspired nudging control; constructed a scalable dataset by sampling the Lorenz-63 attractor with a low-discrepancy strategy and computing expert controls offline.
- In closed-loop filtering experiments, the learned controller accurately reproduced expert behavior but exposed a fundamental tradeoff: strong deterministic control improves state correction but induces particle degeneracy when proposal correction is omitted, establishing that structural compatibility with importance sampling is the primary design constraint.
- IW F24: *Optimal Control in the Filtering of Chaotic Systems Using Adaptive ML Algorithms* (37 pages, January 2025), an initial progress report establishing the HJB-based control formulation, Lorenz-63 dataset generation pipeline using Euler-Maruyama integration, and baseline comparisons across classical filtering methods as groundwork for the thesis.

Leadership

MATRIX Heliophysics Space Mission Architecture Design (MAE 342)

September – December 2025

Project Manager & Team Lead

Princeton University

- Led 16 students through a full-iteration MIDEX-class heliophysics constellation design (MATRIX, \$350M cost cap); recognized as an outstanding leader; most impressive oral presenter; commended at JHU APL site visit.
- Maintained the Gantt Chart and Task Tracker, facilitated weekly cross-subsystem meetings, and reconciled mass, power, data, and cost margins across all eight subsystems; achieved an A+ and 4th-highest individual grade in the course.

Break Through Tech AI, Cornell University

May 2024 – April 2025

Artificial Intelligence & Machine Learning Fellow

Remote

- Built ML pipelines improving model accuracy by **18%** and reducing log loss by **12%** through hyperparameter tuning and regularization.
- Selected for a competitive AI/ML fellowship; completed an industry-partnered challenge project applying end-to-end ML pipelines, from feature engineering through model validation, to a real-world business dataset.

Princeton Rocketry Club

September 2022 – December 2024

Co-President; prev. Systems Engineer, Team Lead, & Professional Development Officer

Princeton University

- Managed 7 cross-functional project teams across **95+ members**; planned a student-led initiative to design and launch a two-stage solid-motor rocket targeting suborbital altitude.
- Taught workshops on CAD (SolidWorks), OpenRocket flight simulation software, composite fabrication, and rocket assembly to new members each semester.

Princeton Autonomous Vehicle Engineering (PAVE)

February – December 2024

Member, Perception & Controls Subteam

Princeton University

- Contributed to the perception stack for an autonomous vehicle platform, working with LiDAR and camera sensor fusion using ROS and OpenCV.
- Implemented and tested object detection and lane-tracking algorithms for real-time deployment on embedded hardware.

Projects

Constellation Analyzer | Three.js r128, HTML/CSS/JS, Orbital Mechanics

May 2026

- Built a browser-based LEO satellite constellation simulator as the capstone for MAE 426 (Rocket Propulsion), exploring the orbital architecture required for a hypothetical 1 GW solar-powered space-based data center.
- Implemented real-time 3D orbital simulation in Three.js r128 (no build step, zero dependencies) with adjustable constellation parameters: number of orbital planes, inclination, altitude, and phase offset; included Starlink, OneWeb, and Polar presets.
- Computed coverage percentage, gap time, orbital period, and ground footprint area entirely client-side, enabling interactive design-space exploration without a backend.

Mt. Rainier Railroad Control System | *6502 Assembly, Arduino C++, Python***Feb – May 2026**

- Designed and built a closed-loop autonomous railroad safety system for the MAE 412 (Microprocessors) capstone, modeling the Mount Rainier Scenic Railroad's fallen-tree hazard scenario on a two-loop physical Atlas flex-track layout.
- Assembled a custom 6502-based vector-board computer from scratch as the supervisory brain of the system, analogous to the 1976 Apple 1 computer and programmed in 6502 assembly; coordinated with an Arduino ATmega328P co-processor over serial and delegated all timing-sensitive I/O to it.
- Programmed the Arduino in C++ to handle the full real-time sense-decide-actuate loop: sampling dual VL53L0X ToF sensors over I2C to detect fallen obstacles at 500 ms intervals, cutting track power via relay upon obstruction, controlling two servo-driven 3D-printed tree obstacles with randomized deployment timing, and switching four motorized turnouts through a trickle-charge circuit.
- Designed all mechanical mounts and tree obstacles in Fusion 360 and fabricated them on BambuLab printers; validated full fail-safe behavior (both tracks de-energized when both hazard zones simultaneously obstructed) across repeated demonstration runs.

MATRIX: Heliophysics Mission Design | *Systems Engineering, Mission Analysis***Jan – May 2026**

- Full-iteration systems engineering design of a MIDEX-class heliophysics constellation (8 spin-stabilized spacecraft, \$350M FY2025 cost cap), studying Bursty Bulk Flows in Earth's magnetotail transition region (6-12 R_E); served as Project Manager across 16 people in 8 mission subsystems.
- Communications, Command & Data Handling:** Conducted the flight software framework trade study (NASA cFS vs. RTEMS vs. VxWorks), selected cFS, and mapped all seven cFE services to MATRIX functions; derived a 1.49 MIPS DPM software load against 11.2 MIPS usable capacity ($7.5\times$ margin); defined the full RS-422 channel allocation across all subsystems to the ATK DPM, verified MA61C-SmallSat to DPM protocol compatibility, and generated the integrated data flow diagram for the two-tier IDPU-to-DPM science data path.
- Operations & Payload:** Authored the end-to-end mission timeline (pre-launch through disposal) decomposed across nine concurrent threads, all spacecraft operational modes and procedures in structured trigger-action-constraint format, a bottom-up BOM cost rollup identifying OPL as the binding cost driver at 5.78% margin, and the cross-subsystem conflict-mitigation analysis resolving four multi-subsystem integration tensions.

StockSwipe | *React, Vite, Node.js, Anthropic SDK, Yahoo Finance, Finnhub API***March 2026**

- Built a mobile-first stock discovery app with Tinder-style swipe mechanics for the AI@Princeton $\times$ Trade[XYZ] Hackathon; users swipe right to invest and left to skip on flippable cards showing live AI analysis, real-time news, and interactive price charts.
- Integrated Claude Haiku and Sonnet via the Anthropic SDK to generate per-stock "hooks" and a holistic AI portfolio analysis with a Risk Radar dashboard, providing natural-language investment reasoning on demand.
- Implemented physics-based swipe gesture animations with Framer Motion, gamified streaks and badges, a social friends feed, and a vertical Reels-style discovery feed to maximize engagement.
- Architected a real-time data layer pulling from Yahoo Finance (pricing/fundamentals), Finnhub (news), and the YouTube Data API (video content), served through an Express/Node.js backend.

Pedestrian Detection Benchmark | *PyTorch, torchvision, OpenCV***Nov – Dec 2025**

- Conducted a controlled architecture-level comparison of four pedestrian detection paradigms (HOG+SVM, Faster R-CNN, DETR, and YOLO+SAM2) evaluated on the PnPLO (Person vs. Person-Like Objects) dataset for the COS 429 (Computer Vision) final project.
- Designed a unified evaluation pipeline computing precision, recall, F1, mAP, false positive breakdowns, PR curves, and confusion matrices across all four models, with a focus on robustness to hard negatives (statues, mannequins, and other visually similar non-pedestrians).
- Analyzed tradeoffs in accuracy, inference speed, and false positive rate across classical, two-stage, transformer-based, and segment-anything architectures; findings documented in a full written report.

RISC-V Processor Simulator | *C++, RV64I, Pipeline Architecture, Cache Simulation***Sep – Dec 2025**

- Built a two-phase RISC-V simulator in C++: a functional simulator executing the full RV64I ISA across 7 explicit stages (Fetch, Decode, Operand Collection, Next-PC Resolution, ALU, Address Generation, Memory Access, and Write-back), validated with 25+ handwritten assembly test cases covering all instruction types.
- Extended the functional simulator into a cycle-accurate 5-stage pipelined processor (IF/ID/EX/MEM/WB), implementing EX $\rightarrow$ ID and MEM $\rightarrow$ ID forwarding paths to resolve data hazards without unnecessary stalls.
- Engineered load-use and load-branch hazard detection with bubble insertion, and branch hazard stall logic, correctly preserving program semantics across concurrent pipeline stages.
- Designed configurable split I-Cache and D-Cache with parameterizable size, block size, set associativity (LRU replacement), and miss latency; modeled cycle-accurate stall propagation on cache misses and exception redirection to a hardware handler.

- Built a full-stack IoT medication adherence system (“NANI”, grandmother in Hindi) for elderly patients, spanning passive IR beam hardware, a bilingual voice-first AI assistant, a cloud data pipeline, and a family-facing iOS app; developed at HackPrinceton Fall 2025.
- Engineered the Raspberry Pi IR beam sensing stack end-to-end (circuit design, Python/FastAPI firmware, and Node.js REST API integration), achieving 98% signal-detection accuracy and sub-150ms event-to-cloud latency.
- Integrated OpenAI Whisper for speech-to-text, GPT-4o-mini for multilingual health Q&A, and OpenAI TTS for voice responses, enabling a culturally sensitive English/Hindi voice assistant accessible to non-English-speaking patients.
- Developed a robust hardware-to-cloud data pipeline with retry logic and local buffering, achieving 99% system uptime; built real-time Care Circle push notifications for family caregivers via the Swift iOS app.

GPU Acceleration of Shap-E via PyTorch MPS | *Python, Metal Performance Shaders*

March – July 2025

- Built a **GPU-acceleration** patch for OpenAI’s Shap-E repository, filed as a Pull Request, enabling image-to-3D **inference** on Apple Silicon and reducing inference time from **4+ hours to under 4 minutes**; rewrote core **tensor-indexing** logic for GPU-based parallel execution while preserving CUDA/CPU compatibility.
- Validated functional parity across CUDA and MPS backends by analyzing 3D mesh outputs from 25+ diffusion sampling runs, confirming no regression in generation quality.

Autonomous Obstacle-Avoiding Drone | *Python, OpenCV, CrazyFlie, Kalman Filter*

September – December 2024

- Built a fully autonomous CrazyFlie drone for MAE 345 (Robotics): real-time HSV obstacle detection, contour-based avoidance, and closed-loop waypoint following via onboard Kalman filter; selected by Prof. Nagpal as a reference implementation for future cohorts.
- Implemented real-time computer vision entirely in OpenCV: designed dual-range HSV masking to robustly detect red obstacles under varying lighting conditions, computed contour-based proximity zones (three threat levels thresholded at 5K, 10K, and 23K px²), and built a separate blue-channel bounding-box tracker for target acquisition and proportional approach control.
- Engineered the full autonomous decision loop: pre-flight lateral sweep to select the least-obstructed starting lane, real-time obstacle avoidance with stateful boundary tracking via a global direction flag to prevent feedback oscillation, and a vision-based landing controller triggered when the target bounding box exceeded 4% of frame area.

Unwrapping Customer Delight | *Python, NumPyro, JAX, StatsModels, Pandas*

August – December 2024

- Developed a Bayesian Regression Discontinuity Design (RDD) to evaluate the causal ROI of surprise gift campaigns on future customer spending, using 50,000+ anonymized transaction records from The Estée Lauder Companies.
- Applied MCMC sampling (NumPyro/JAX) to fit a Bayesian regression model with informative priors, reducing credible interval widths by 15% and providing actionable posterior distributions over gift-driven revenue lift for marketing budget allocation.

Airbnb Price Prediction | *Python, Pandas, scikit-learn, Seaborn, Matplotlib*

July – August 2024

- Built an end-to-end regression pipeline to predict New York City Airbnb listing prices from 50,000+ structured records; performed EDA, outlier removal, and feature engineering (one-hot encoding, log transforms).
- Trained and compared Ridge Regression, Decision Tree, and Random Forest models; GridSearchCV hyperparameter tuning yielded a 9.5% improvement in R^2 and a 10% reduction in RMSE, with Random Forest achieving the best overall performance.

ML-Based Fraud Detection System | *Java, Clustering, Decision Stump, Kruskal MST*

April – May 2024

- Implemented a fraud detection system combining graph-based clustering (Kruskal’s MST) with a boosted decision-stump weak learner to classify anomalous financial transactions, achieving 98% accuracy on held-out test data.
- Analyzed precision/recall tradeoffs to tune the decision threshold for a low false-negative operating point, appropriate for real-world fraud detection where missed fraud is costlier than false alarms.

NeuroBoost: Cognitive Function Monitor | *Python, TensorFlow, Keras, NumPy, Bootstrap*

March – May 2024

- Developed a proof-of-concept real-time cognitive monitoring system using biosensors measuring galvanic skin response (GSR) and heart rate variability (HRV) as proxies for cognitive load and fatigue.
- Designed and trained a hybrid CNN-RNN architecture in TensorFlow/Keras to classify cognitive state (baseline, load, fatigue) from multivariate time-series biosensor data; built a live Bootstrap/JavaScript dashboard to stream sensor readings and model predictions in real time.

Genetic Algorithm Orbit Optimizer | *Python, MATLAB, Orbital Mechanics*

June – August 2023

- Designed and implemented a multi-objective genetic algorithm (selection, crossover, mutation, Pareto front ranking) for optimizing CubeSat orbital parameters subject to mission coverage, power, and delta-v constraints for the Hellenic Space Dawn mission.
- Reduced orbit design computation time by 30% versus exhaustive grid search across 1,000+ candidate scenarios; results were incorporated into an ESA-reviewed mission Design Review.